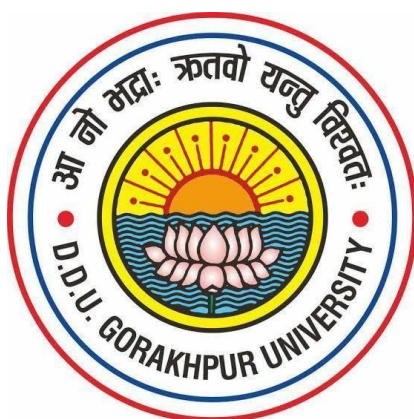


**INSTITUTE OF ENGINEERING AND TECHNOLOGY
DEEN DAYAL UPADHYAYA GORAKHPUR UNIVERSITY,
GORAKHPUR**



**COURSE STRUCTURE
&
SYLLABUS**

Master of Technology (M. Tech.)

Based

on

AICTE MODEL CURRICULUM

[Effective from the Session: 2025-26]



INSTITUTE OF ENGINEERING AND TECHNOLOGY
Deen Dayal Upadhyaya Gorakhpur University, Gorakhpur

Curriculum for Master of Technology
Course Structure & Semester-wise Credit Distribution

Credit Distribution (Course Type) in M. Tech. Program		
S.No.	Course Type	Credit
1	PCC	44
2	PEC	09
3	Seminar	02
4	Dissertation	33 (15+18)
Total Credit		88

Abbreviation:

PCC: Professional Core Course

PEC: Professional Elective Course



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Professional Core Course (PCC)			
S.No.	Course Code	Course Title	Credit
1.	MCSE-101	Foundation of Computer Science	3+0
2.	MCSE-102	Research Process and Methodology	3+0
3.	MCSE-103	Advanced Algorithm	3+0
4.	MCSE-104	Data Mining Techniques	3+0
5.	MCSE-105	Soft Computing Techniques	3+0
6.	MCSE-106	Cryptography and Network Security	3+0
7.	MCSE-201	Multi Core Architecture and Programming	3+0
8.	MCSE-202	Wireless Mobile Networks	3+0
9.	MCSE-203	Blockchain and Its Applications	3+0
10.	MCSE-204	Advanced Distributed Systems	3+0
11.	MCSE-205	Big Data and Its Applications	3+0
12.	MCSE-206	Advanced Computer Networks	3+0
13.	MCSE-151	Foundation of Computer Science Lab	0+2
14.	MCSE-153	Advanced Algorithm Lab	0+2
15.	MCSE-252	Wireless Mobile Networks Lab	0+2
16.	MCSE-254	Advanced Distributed Systems Lab	0+2
Total			44

Professional Elective Course (PEC)			
S.No.	Semester	PEC-Elective	Credit
1.	III	Elective-I	3+0
2.	III	Elective-I	3+0
3.	IV	Elective-II	3+0
Total			09
Student has to select unique course(s) either from Elective-I or from Elective-II in the semester. Repetition of courses is not allowed.			

Project Work, Seminar and Internship/Industrial Training in Industry or Elsewhere			
S.No.	Course Code	Course Title	Credit
1.	MCSES-301	Seminar	0+2
2.	MCS-301	Dissertation-I	0+15
3.	MCS-401	Dissertation-II	0+18
Total			35



INSTITUTE OF ENGINEERING AND TECHNOLOGY

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AEC and SEC Offered by the University for Implementation of NEP2020 (University Mandatory Course)

As per NEP2020, year wise credit requirements for the award of “Advanced Diploma in Computer Science & Engineering”, “Post Graduate Diploma in Computer Science & Engineering”, and “Master of Technology” are given below:

First Year: After earning 44 credits in first year (22 credits in first semester and 22 credits in second semester), student will be eligible for “**Advanced Diploma in Computer Science & Engineering**”.

Second Year (3rd Semester): After earning 67 credits (44 credits from first year and 23 credits in second year, i.e. 23 credits in third semester), student will be eligible for the award of “**Post Graduate Diploma in Computer Science & Engineering**”.

Second Year (4th Semester): After earning 88 credits (44 credits from first year and 44 credits in second year), student will be eligible for the degree of “**Master of Technology**”.

Table1: Year wise credit requirement for award of Advanced Diploma in Computer Science & Engineering, Post Graduate Diploma in Computer Science & Engineering, and Master of Technology under the light of NEP2020.

After Year	Credit Requirement	Eligibility of
1 st Year	44	Advanced Diploma in Computer Science & Engineering
2 nd Year (3 rd Semester)	67	Post Graduate Diploma in Computer Science & Engineering
2 nd Year (4 th Semester)	88	Master of Technology



INSTITUTE OF ENGINEERING AND TECHNOLOGY
Deen Dayal Upadhyaya Gorakhpur University, Gorakhpur

Professional Elective Courses (from Department of CSE) -1

S.No.	Course Code	Course Name	Credit
1.	MCSEE-101	Sensor Network	3+0
2.	MCSEE-102	Software Testing and Auditing	3+0
3.	MCSEE-103	Real Time Systems	3+0
4.	MCSEE-104	Genetic Algorithms	3+0
5.	MCSEE-105	Neural Networks	3+0
6.	MCSEE-106	Data Warehousing & Data Mining	3+0

Note:

- Students may take subjects from Elective-1 (from SWAYAM) and/or Elective-1 (from Department of CSE)
- Repetition of courses is not allowed.
- Execution of the courses offered by the department depends on the availability of faculty persons.



INSTITUTE OF ENGINEERING AND TECHNOLOGY

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Professional Elective Courses (from SWAYAM) -1

Duration: 12 Weeks, Credit: 3

S.No.	Course Code	Course Id	Course Name	SME Name	Institute	Link
1.	MCSEE-107	noc23-cs127	Cyber Security and Privacy	Prof. Saji K Mathew	IITM	https://onlinecourses.nptel.ac.in/noc23_cs127/ preview
2.	MCSEE-108	noc23-cs77	Computer Vision	Prof. Jayanta Mukhopadhyay	IITKGP	https://onlinecourses.nptel.ac.in/noc23_cs77/ p review
3.	MCSEE-109	noc23-cs89	Cloud Computing	Prof. Soumya Kanti Ghosh	IITKGP	https://onlinecourses.nptel.ac.in/noc23_cs89/ p review
4.	MCSEE-110	noc23-cs114	C-BasedVLSI Design	Prof. Chandan Karfa	IITG	https://onlinecourses.nptel.ac.in/noc23_cs114/ preview
5.	MCSEE-111	noc23-cs92	Artificial Intelligence : Search Methods For Problem Solving	Prof. Deepak Khemani	IITM	https://onlinecourses.nptel.ac.in/noc23_cs92/ p review
6.	MCSEE-112	noc23-cs102	Parameterized Algorithms	Prof. Neeldhara Misra	IIT Gandhinagar	https://onlinecourses.nptel.ac.in/noc23_cs102/ preview
7.	MCSEE-113	noc23-cs107	Social Networks	Prof. Sudarshan Iyengar Prof. Yayati Gupta	IIT Ropar	https://onlinecourses.nptel.ac.in/noc23_cs107/ preview
8.	MCSEE-114	noc23-cs111	Computational Complexity	Prof. Subrahmanyam Kalyanasundaram	IIT Hyderabad	https://onlinecourses.nptel.ac.in/noc23_cs111/ preview
9.	MCSEE-115	noc23-cs68	Advanced Distributed Systems	Prof. Smruti Ranjan Sarangi	IITD	https://onlinecourses.nptel.ac.in/noc23_cs68/ p review
10.	MCSEE-116	noc23-cs84	Statistical LearningFor Reliability Analysis	Prof. Monalisa Sarma	IITKGP	https://onlinecourses.nptel.ac.in/noc23_cs84/ p review
11.	MCSEE-117	noc23-cs75	Ethical Hacking	Prof. Indranil Sengupta	IITKGP	https://onlinecourses.nptel.ac.in/noc23_cs75/ p review
12.	MCSEE-118	noc23-cs91	Software Testing	Prof. Meenakshi D'Souza	IIITB	https://onlinecourses.nptel.ac.in/noc23_cs91/ p review
13.	MCSEE-119	noc23-cs126	Deep Learning For Computer Vision	Prof. Vineeth N Balasubramanian	IITH	https://onlinecourses.nptel.ac.in/noc23_cs126/ preview
14.	MCSEE-120	noc23-cs69	Privacy And In Online Soci al Media	Prof. Ponnurangam Kumaraguru	IIITH	https://onlinecourses.nptel.ac.in/noc23_cs69/ p review

Note: In case of changes in the above subjects , Dean/Director/ HoD will issue separate list of subjects.



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Professional Elective Courses (from Department of CSE) -2

S.No.	Course Code	Course Name	Credit
1.	MCSEE-201	Machine Learning	3+0
2.	MCSEE-202	High Performance Networking	3+0
3.	MCSEE-203	Big Data Analytics	3+0
4.	MCSEE-204	Software Metrics and Quality Assurance	3+0

Note:

- Students may take subjects from Elective-2 (from SWAYAM) and/or Elective-2 (from Department of CSE)
- Repetition of courses is not allowed.
- Execution of the courses offered by the department depends on the availability of faculty persons.



INSTITUTE OF ENGINEERING AND TECHNOLOGY

Deen Dayal Upadhyaya Gorakhpur University, Gorakhpur

Professional Elective Courses (from SWAYAM) -2

Duration: 12 Weeks, Credit: 3

S.No.	Course Code	Course Id	Course Name	SME Name	Institute	Link
1.	MCSEE-205	noc23-cs07	Advanced Computer Architecture	Prof. Smruti R.Sarang	IITD	https://onlinecourses.nptel.ac.in/noc23_cs07
2.	MCSE- 206	noc23-cs59	Hardware Security	Prof. Debdeep Mukhopadhyay	IITKGP	https://onlinecourses.nptel.ac.in/noc23_cs59
3.	MCSEE-207	noc23-cs61	GPU Architectures And Programming	Prof. Soumyajit Dey	IITKGP	https://onlinecourses.nptel.ac.in/noc23_cs61
4.	MCSEE-208	noc23-cs62	Foundations of Cyber Physical Systems	Prof. Soumyajit Dey	IITKGP	https://onlinecourses.nptel.ac.in/noc23_cs62
5.	MCSEE- 209	noc23-cs63	Selected Topics in Algorithms	Prof. Palash Dey	IITKGP	https://onlinecourses.nptel.ac.in/noc23_cs63
6.	MCSEE- 210	noc23-cs36	Affective Computing	Prof. Jainendra Shukla Prof. Abhinav Dhall	IIITD	https://onlinecourses.nptel.ac.in/noc23_cs36
7.	MCSEE-211	noc23-cs51	Introduction To Internet Of Things	Prof. Sudip Misra	IITKGP	https://onlinecourses.nptel.ac.in/noc23_cs51
8.	MCSEE- 212	noc23-cs08	Data Analytics with Python	Prof. A. Ramesh	IITR	https://onlinecourses.nptel.ac.in/noc23_cs08
9.	MCSEE- 213	noc23-cs03	Foundations of Cryptography	Prof. Ashish Choudhury	IIIT Bangalore	https://onlinecourses.nptel.ac.in/noc23_cs03
10.	MCSEE -214	noc23-cs23	Reinforcement Learning	Prof. Balaraman Ravindran	IITM	https://onlinecourses.nptel.ac.in/noc23_cs23
11.	MCSEE- 215	noc23-cs24	Deep Learning	Prof. Sudarshan Iyengar	IIT Ropar	https://onlinecourses.nptel.ac.in/noc23_cs24
12.	MCSEE- 216	noc23-cs45	Natural Language Processing	Prof. Pawan Goyal	IITKGP	https://onlinecourses.nptel.ac.in/noc23_cs45

Note: In case of changes in the above subjects , Dean/Director/ HoD will issue separate list of subjects.



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Credit Distribution (Semester wise) (Minimum Common Programme)

S.No.	Semester Number	Credit
1	Semester I	22
2	Semester II	22
3	Semester III	23
4	Semester IV	21
Total Credit		88



Course Structure



INSTITUTE OF ENGINEERING AND TECHNOLOGY
Deen Dayal Upadhyaya Gorakhpur University, Gorakhpur

Branch/Course: M.Tech.(Computer Science & Engineering)

Semester-I (First Year) Curriculum

S. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Core Course	MCSE-101	Foundation of Computer Science	3+0
2.	Professional Core Course	MCSE-102	Research Process and Methodology	3+0
3.	Professional Core Course	MCSE-103	Advanced Algorithm	3+0
4.	Professional Core Course	MCSE-104	Data Mining Techniques	3+0
5.	Professional Core Course	MCSE-105	Soft Computing Techniques	3+0
6.	Professional Core Course	MCSE-106	Cryptography and Network Security	3+0
7.	Professional Core Course	MCSE-151	Foundation of computer Science Lab	0+2
8.	Professional Core Course	MCSE-153	Advanced Algorithm Lab	0+2
Total Credit				22

Semester-II (First Year) Curriculum

S. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Core Course	MCSE-201	Multi Core Architecture and Programming	3+0
2.	Professional Core Course	MCSE-202	Wireless Mobile Networks	3+0
3.	Professional Core Course	MCSE-203	Blockchain and Its Applications	3+0
4.	Professional Core Course	MCSE-204	Advanced Distributed Systems	3+0
5.	Professional Core Course	MCSE-205	Big Data and Its Applications	3+0
6.	Professional Core Course	MCSE-206	Advanced Computer Networks	3+0
7.	Professional Core Course	MCSE-252	Wireless Mobile Networks Lab	0+2
8.	Professional Core Course	MCSE-254	Advanced Distributed Systems Lab	0+2
Total Credit				22



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Branch/Course: M.Tech.(Computer Science & Engineering) Semester-III (Second Year) Curriculum

S. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Elective Course	MCSEE-***	Elective-I (either from Department or from SWAYAM)	3+0
2.	Professional Elective Course	MCSEE-***	Elective-I (either from Department or from SWAYAM)	3+0
3.	Seminar	MCSES-301	Seminar	0+2
4.	Project	MCSED-301	Dissertation-I	0+15
Total Credit				23

Semester-IV (Second Year) Curriculum

S. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Elective Course	MCSEE-***	Elective-II (either from Department or from SWAYAM)	3+0
2.	Project	MCSED-401	Dissertation-II	0+18
Total Credit				21

Master of Technology

Foundation of Computer Science (MCSE-101)

Credit: 3+0

Course Objective: Foundation of Computer Science provides fundamental knowledge of computing principles, algorithms, data structures, and problem-solving techniques. It aims to develop logical thinking and programming skills essential for software development and system analysis. to provide fundamental knowledge of computing principles, algorithms, data structures, and problem-solving techniques. It aims to develop logical thinking and programming skills essential for software development and system analysis.

Syllabus

1. **Data Structure:** List, Stack, Queue, Tree, Hash Table, Graph, Search and Sorting Algorithms.
2. **Operating System:** CPU Scheduling Algorithm, Synchronization Techniques, Paging and Segmentation, Virtual Memory.
3. **Automata Theory:** Finite Automata, Regular Expression, Context Free Grammar, Push Down Automata, Turing Machine, P and NP Class.
4. **Distributed Database:** Distributed Database Concepts, Types of Distributed Database Systems, Distributed Database Architectures, Data Fragmentation, Replication and Allocation Techniques for Distributed Database systems, Query Processing and Optimization in Distributed Databases, Overview of Transaction Management, Concurrency Control and Recovery in Distributed Databases, Distributed Catalog Management.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To Gain knowledge of essential data structures like lists etc, along with searching and sorting techniques to optimize problem-solving.

CO2: To comprehend key OS concepts, including scheduling algorithms and to ensuring efficient resource utilization.

CO3: To develop an understanding of formal language theory.

CO4: To learn distributed database architectures for efficient data distribution.

References:

1. Hopcroft & Ullman, "Introduction to Automata Theory, Languages, and Computation", Narosa Publishing House.
2. Aho, Hopcroft and Ullman, "Data Structures and Algorithms", Addison Wesley.
3. Dhamdhare, "Operating Systems", Tata McGraw Hill.
4. Aho, Ullman and Sethi, "Compiler Design", Addison Wesley.
5. Ramez Elmasri, Shamkant Navathe: Fundamentals of Database Systems, Fifth Edition, Pearson Education, 2007. 6.Raghu Ramakrishnan, Johannes Gehrke : Database Management Systems, Tata McGraw-Hill.
6. Alexis Lcon, Mathews Leon, "Database Management Systems
7. C.J. Date : An Introduction to Database Systems, Eighth Edition, Pearson Education.
8. Abraham Silberschatz, Henry F. Korth, S.Sudarshan: Database System Concepts, Tata McGraw-Hill.

Master of Technology

Research Process and Methodology (MCSE-102)

Credit: 3+ 0

Course Objective: Students will be able to clearly define the research question, guide the data collection and analysis methods, ensure the study is conducted systematically, and ultimately achieve the research goals by providing reliable and valid results; this involves specifying the research design, selecting appropriate methods (qualitative, quantitative, or mixed), determining sampling strategies, and outlining data analysis techniques to answer the research questions effectively.

Syllabus

1. Introduction Meaning, objectives and motivations in research, Characteristics and limitations of research – Components of research work - Criteria of good research, Research process – Types of Research, Fundamental, Pure or Theoretical Research –Applied Research Descriptive Research – Evaluation Research –Experimental Research – Survey Research – Qualitative Research – Quantitative Research – Historical Research.
2. Research Design Research Design – definition – essentials and types of research design – errors and types of errors in research design. Research problem: Selecting and analyzing the research problem – problem statement formulation – formulation of hypothesis. Literature review: purpose, sources, and importance - literature review procedure. Objectives: Learning Objectives; Definitions; Formulation of the research objectives.
3. Measurement. Scaling and Sampling Variables in Research – Measurement and scaling – Different scales, Data Collection methods – primary and secondary data – Construction of questionnaire and instrument – validation of instruments. Sample size determination - Sample design and sampling techniques. Data Analysis and Tools Processing of Data: Editing of Data – Coding of Data – Classification of Data – Statistical Series. Qualitative vs Quantitative data analyses ,Dispersion, correlation and Regression, Chi-square test: Applications, Steps, characteristics, limitations, Analysis of Variance and Covariance, Factor analysis – Discriminant analysis – cluster analysis – multiple regression and correlation – multidimensional scaling – Conjoint Analysis - Application of statistical software for data analysis.
4. Research Report Writing Research report – Different types – Contents of report executive summary – chapterization – contents of chapter – report writing – the role of audience – readability – comprehension – tone – final proof – report format – title of the report – Ethical issues in research: Code of Ethics in Research – Ethics and Research Process – Importance of Ethics in Research.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To Gain knowledge of research concepts along with understanding various research methodologies,

CO2: To learn to design effective research studies, identify and analyze research problems, formulate hypotheses, and conduct thorough literature reviews to establish a strong research foundation.

CO3: To understand different research variables, data collection methods, questionnaire design, sampling techniques, and measurement scales to ensure data accuracy and validity..

CO4: To utilize statistical tools for data processing, analysis (correlation, regression, ANOVA, factor analysis), and interpretation, while effectively writing research reports with ethical considerations.

References:

1. C. R. Kothari, Gaurav Garg, Research Methodology Methods and Techniques , New Age International publishers, Third Edition.
2. Ranjit Kumar, Research Methodology: A Step-by-Step Guide for Beginners, 2nd Edition, SAGE, 2005
3. Business Research Methods – Donald Cooper & Pamela Schindler, TMGH, 9th edition
4. Creswell, John W. Research design: Qualitative, quantitative, and mixed methods approaches. Sage publications, 2013.

Master of Technology

Advanced Algorithm (MCSE-103)

Credit: 3+ 0

Course Objective: Students will be able to solve complex problems and it can be used in many fields, including machine learning, operations research, and bridge construction.

Syllabus

1. Algorithm Fundamentals: Basic Concept, Analysis of Algorithm, Growth of Functions, Master's Theorem. Analysis of sorting Algorithms: Overview, Merge sort, Quick sort, Heap sort, radix sort.
2. Advance Data Structure: Red-Black Trees, B/B+ Trees Parallel Algorithm: Performance Measures of Parallel, Algorithms, Parallel Merging/Sorting Algorithms on CREW/EREW, Parallel searching algorithms.
3. Advance Design and Analysis Techniques: Dynamic Programming, Greedy Algorithms, Branch and Bound, Back Tracking.
4. Graph Algorithm: DFS and BFS algorithm, Minimum Spanning Trees, Single Source Shortest Path, All Pair Shortest Path, Maximum Flow Problem, NP Complete Problem

Course Outcomes (CO): At the end of this course students will be able:

CO1: To understand fundamental algorithm concepts and assess the efficiency of sorting algorithms such as Merge Sort, Quick Sort, Heap Sort, and Radix Sort.

CO2: To gain proficiency in Red-Black Trees, B/B+ Trees, and parallel algorithm techniques, including parallel sorting, merging, and searching on CREW/EREW models.

CO3: To develop efficient solutions for complex computational problems.

CO4: To Implement BFS and DFS for graph traversal, explore NP-complete problems, and understand the computational complexity of intractable problems.

References:

1. Coreman, Rivest, Lisserson, "Algorithm", PHI.
2. Basse, "Computer Algorithms: Introduction to Design & Analysis", Addison Wesley.
3. Horowitz, Sahani, and Rajasekaran "Fundamental of Computer Algorithms", Universities Press.

Master of Technology

Data Mining Techniques (MCSE-104)

Credit: 3+0

Objectives: Students will be able to extract meaningful patterns, insights, and correlations from large datasets to make informed predictions or decisions, often by identifying hidden relationships within the data that can be used to understand trends, behaviors, or future outcomes across various domains like business, healthcare, and finance.

Syllabus

1. Data Mining Techniques- Process with a typical collection of data, Big Data, and data visualisation using data mining software, Data Mining Methods as Tools - Data Mining Memory, Based Reasoning Methods, Algorithms with simulated data based on real-world applications
2. Data Stream Mining, Time Series Mining, Text Mining, Data Stream Clustering, and Big Data Mining are all terms for the same thing.
3. Market Basket Analysis - Fuzzy Data Mining, Fuzzy Decision Tree, Applications of the Fuzzy Association Rule, Support Vector Machines, Genetic Algorithms, Rough Sets
4. Graph Mining, Social Network Mining, Web Mining, Web Usage Mining, Privacy-Preserving Data Mining, and Recommender Systems are examples of social computing.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To learn fundamental data mining processes, big data handling, and data visualization using modern software tools.

CO2: To explore techniques like data stream mining, time series mining, text mining, and clustering for large-scale data analysis.

CO3: To implement market basket analysis, fuzzy logic, support vector machines, genetic algorithms, and rough sets for pattern discovery and decision-making.

CO4: To gain insights into graph mining, social network analysis, web mining, recommender systems, and privacy-preserving data mining for real-world applications.

References:

1. Steinbach and Vipin Kumar, Introduction to Data Mining, Tan, , Pearson Education, 2016
2. Pei, Han and Kamber Data Mining: Concepts and Techniques, , Elsevier, 2011

Master of Technology

Soft Computing Techniques (MCSE-105)

Credit: 3+0

Course Objectives: To develop intelligent systems capable of solving real-world problems by leveraging tolerance for imprecision, uncertainty, partial truth, and approximation, essentially mimicking the human decision-making process to achieve robust and cost-effective solutions, often in situations where precise mathematical modelling is difficult or impossible; key components include fuzzy logic, neural networks, evolutionary computing, and probabilistic reasoning.

Syllabus

1. Fuzzy sets, Membership functions, Basic operation, Fuzzy relations, Defuzzification, Fuzzy inference, Fuzzy rule based system, Working principle of Genetic algorithm, Population set, Binary encoding and decoding. Selection (roulette wheel selection and other methods), Cross over techniques, Mutation,
2. Problems of Genetic Algorithms. Multi objective GAs, Concepts on non-domination, Crowding distance operator, Ranking, and applications.
3. Rough Sets, Lower and upper approximations, Accuracy of Approximations, Discernibility matrix, Discernibility Function, Reduct and core.
4. Rule Generations and application of rough set theory, Hybridization of soft computing tools like fuzzy-rough.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To learn fuzzy sets, membership functions, defuzzification, and genetic algorithm principles like selection, crossover, and mutation.

CO2: To explore challenges, non-domination concepts, ranking, and real-world applications of genetic algorithms.

CO3: To understand lower and upper approximations, discernibility matrices, reducts, and core concepts for data analysis.

CO4: To develop rule generation methods and hybridize fuzzy and rough set techniques for advanced problem-solving.

References:

1. G.J.Klir and B.Yuan, Fuzzy Sets and Fuzzy Logic: Theory and Applications, Prentice-Hall, 1995.
2. M.Mitchell, An Introduction to Genetic Algorithms, MIT Press, 2000
3. Dario Floreano, Claudio Mattiussi, "Bio-Inspired Artificial Intelligence: Theories, Methods and Technologies", MIT Press, 2008.
4. Eberhart, E. and Y. Shi., "Computational Intelligence: Concepts and Implementations", Morgan Kaufmann, San Diego, 2007.
5. D. E. Goldberg, "Genetic algorithms in search, optimization, and machine learning", AddisonWesley, 1989
6. R. C. Ebelhart et al., "Swarm Intelligence", Morgan Kaufmann, 2001. • M. Dorigo and T. Stutzle, "Ant Colony Optimization", A Bradford Book, 2004.
7. Leandro Nunes De Castro, Fernando Jose Von Zuben, "Recent Developments in Biologically

Inspired Computing", Idea Group Publishing, 2005.

8. Kenneth DeJong, Evolutionary Computation A Unified Approach, 2006.
9. Gusz Eiben and Jim Smith, Introduction to Evolutionary Computing, 2007.
10. Simon O. Haykin, "Neural Networks and Learning Machines", Third Edition, Prentice Hall, 2008.
11. A.E. Elben and J. E. Smith, "Introduction to Evolutionary Computing", Springer, 2010.

Master of Technology

Cryptography and Network Security (MCSE-106)

Credits: 3+0

Course Objectives: Student will be able to protect data transmitted across networks by ensuring its confidentiality, integrity, and availability through the use of encryption algorithms, authentication mechanisms, and other security protocols, effectively preventing unauthorized access, modification, or disruption of information.

1. Introduction to Cryptography and Block Ciphers, Introduction to security attacks - services and mechanism - introduction to cryptography - Conventional Encryption: Conventional encryption model - classical encryption techniques - substitution ciphers and transposition ciphers - cryptanalysis - steganography - stream and block ciphers - Modern Block Ciphers: Block ciphers principals - Shannon's theory of confusion and diffusion - Feistel structure - data encryption standard (DES) - strength of DES - differential and linear cryptanalysis of DES - block cipher modes of operations - triple DES - AES.
2. Confidentiality and Modular Arithmetic Confidentiality using conventional encryption - traffic confidentiality - key distribution - random number generation - Introduction to group - ring and field - prime and relative prime numbers - modular arithmetic - Fermat's and Euler's theorem - primality testing - Euclid's Algorithm - Chinese Remainder theorem - discrete algorithms.
3. Public key cryptography and Authentication requirements Principles of public key crypto systems - RSA algorithm - security of RSA - key management - Diffie-Hellman key exchange algorithm - introductory idea of Elliptic curve cryptography - ElGamal encryption - Message Authentication and Hash Function: Authentication requirements - authentication functions - message authentication code - hash functions - birthday attacks - security of hash functions and MACS.
4. Integrity checks and Authentication algorithms, MD5 message digest algorithm - Secure hash algorithm (SHA) Digital Signatures: Digital Signatures - authentication protocols - digital signature standards (DSS) - proof of digital signature algorithm - Authentication Applications: Kerberos and X.509 - directory authentication service - electronic mail security - Pretty Good Privacy (PGP) - S/MIME.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To comprehend classical and modern encryption methods, including DES and AES, and their cryptanalysis.

CO2: To utilize number theory and modular arithmetic principles in cryptographic contexts.

CO3: To employ algorithms like RSA, Diffie-Hellman, and elliptic curve cryptography for secure communications.

CO4: To Use hash functions, digital signatures, and authentication protocols to maintain data security.

References:

1. W. Mao, "Modern Cryptography – Theory and Practice", Pearson Education.
2. Charles P. Pfleeger, Shari Lawrence Pfleeger – Security in computing – Prentice Hall of India.

Master of Technology

Foundation of Computer Science Lab (MCSE-151)

Credits: 0+2

1. Implementation of List, Stack, Queue, Tree, Hash Table, Graph, Search and Sorting Algorithms
2. Implementation of Scheduling Algorithm, Synchronization Technique, Paging and Segmentation, Virtual Memory.
3. Implementation of Finite Automata, Regular Expression, Context Free Grammar, Push Down Automata, Turing Machine, P and NP Class.
4. Implementation of – Phases, Lexical Analysis, Syntax and Semantic Analysis, Top Down and Bottom Up Parsing Techniques.

Note: Implementation may be done in C, C++, Java and Python. Experiments may be added or removed in justified manner.

1. Write Java programs that use both recursive and non-recursive functions for implementing the following searching methods: a) Linear search b) Binary search
2. Write a Java program to implement all the functions of a dictionary (ADT) using Hashing.
3. Write a Java program to implement Dijkstra's algorithm for Single source shortest path problem.
4. Write Java programs that use recursive and non-recursive functions to traverse the given binary tree in a) Preorder b) Inorder c) Postorder.
5. Write Java programs for the implementation of bfs and dfs for a given graph.
6. Write Java programs for implementing the following sorting methods: a) Bubble sort b) Insertion sort c) Quick sort d) Merge sort e) Heap sort f) Radix sort g) Binary tree sort
7. Write a Java program to perform the following operations: a) Insertion into a B-tree b) Searching in a B-tree
8. Write a Java program that implements Kruskal's algorithm to generate minimum cost spanning tree.
9. Write a Java program that implements KMP algorithm for pattern matching.

Note: Implementation may be done in C, C++, and Python also. Experiments may be added or removed in justified manner.

Master of Technology

Multi Core Architecture and Programming (MCSE-201)

Credit: 3+0

Course Objective: Students will be able to improve performance and multitasking capabilities by using multiple processing cores on a single chip.

Syllabus

1. Introduction to Multi-core Architecture: Motivation for Concurrency in software, Parallel Computing Platforms, Parallel Computing in Microprocessors, Differentiating Multi-core Architectures from Hyper- Threading Technology, Multi-threading on Single-Core versus MultiCore Platforms Understanding Performance, Amdahl's Law, Growing Returns: Gustafson's Law. System Overview of Threading: Defining Threads, System View of Threads, Threading above the Operating System, Threads inside the OS, Threads inside the Hardware, What Happens When a Thread Is Created, Application Programming Models and Threading, Virtual Environment: VMs and Platforms, Runtime Virtualization, System Virtualization.
2. Fundamental Concepts of Parallel Programming: Designing for Threads, Task Decomposition, Data Decomposition, Data Flow Decomposition, Implications of Different Decompositions, Challenges You'll Face, Parallel Programming Patterns, A Motivating Problem: Error Diffusion, Analysis of the Error Diffusion Algorithm, An Alternate Approach: Parallel Error Diffusion, Other Alternatives.
3. Threading and Parallel Programming Constructs: Synchronization, Critical Sections, Deadlock, Synchronization Primitives, Semaphores, Locks, Condition Variables, Messages, Flow Control-based Concepts, Fence, Barrier, Implementation-dependent Threading Features. Threading APIs : Threading APIs for Microsoft Windows, Win32/MFC Thread APIs, Threading APIs for Microsoft .NET Framework, Creating Threads, Managing Threads, Thread Pools, Thread Synchronization, POSIX Threads, Creating Threads, Managing Threads, Thread Synchronization, Signaling, Compilation and Linking.
4. Open MP: A Portable Solution for Threading: Challenges in Threading a Loop, Loop- carried Dependence, Data-race Conditions, Managing Shared and Private Data, Loop Scheduling and Portioning, Effective Use of Reductions, Minimizing Threading Overhead, Work-sharing Sections, Performance-oriented Programming, Using Barrier and No wait, Interleaving Singlethread and Multi-thread Execution, Data Copy-in and Copy-out, Protecting Updates of Shared Variables, Intel Task queuing Extension to OpenMP, OpenMP Library Functions, OpenMP Environment Variables, Compilation, Debugging, performance.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To comprehend the motivation, design, and performance considerations of multi-core processors,

CO2: To design and implement parallel programs using task and data decomposition strategies, addressing challenges such as synchronization and data dependencies.

CO3: To employ threading mechanisms, synchronization primitives, and threading APIs (e.g., POSIX threads, OpenMP) to develop efficient concurrent applications.

CO4: To effectively use OpenMP directives, manage shared and private data, and optimize performance in multi-threaded environments.

References:

1. Shameem Akhter and Jason Roberts ,Multicore Programming , Increased Performance through Software Multi-threading by , Intel Press , 2006
2. Sashi Kumar ,Introduction to Parallel Processing , ,PHI
3. Wilkinson ,Parallel Programming, , Pearson
4. Rajaraman ,Elements of Parallel Computing, ,PHI

Master of Technology

Wireless Mobile Networks (MCSE-202)

Credits: 3+0

Course Objective: Students will be able to enable seamless communication between devices without the need for physical cables, allowing for mobility and data transmission across various locations, primarily achieved through technologies like cellular networks, Wi-Fi, and Bluetooth, which facilitate connectivity while users are on the move; key goals include providing reliable access to the internet, making calls, sending data, and utilizing location-based services, all while managing signal strength and network congestion across different geographic areas.

Syllabus

1. Introduction to Wireless and Mobile Networks, Wireless Transmission: Signals, Antennas, Signal Propagation, Multiplexing, Modulation, Spread Spectrum, Wireless Medium Access Control: Common Problems, SDMA, FDMA, TDMA, CDMA, Wireless Telecommunications Systems: GSM, DECT, TETRA, UMTS, IMT-2000, LTE
2. Satellite Systems: Introduction, Deficiencies of existing GEO/MEO/LEO Satellite Systems, Satellite Architectures, Satellite Routing, Satellite Channel Access, Satellite Handover, High Altitude Platforms, and Applications. Wireless LAN: IEEE 802.11, Bluetooth, RFID, Security issues. Mobile Network Layer I: Problems of IP in Wireless, Principles behind Mobile IP, Problems, Security issues, DHCP.
3. Mobile Network Layer II: Routing in Ad-hoc Networks, Wireless Sensor Networks, Mobile Transport Layer: Effects of mobility and wireless transmissions on reliable transport protocols such as TCP.
4. Support for Mobility: File Systems, databases, WWW and Mobility, WAP, Application layer for mobile networks.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To grasp the fundamentals of wireless transmission and multiple access methods such as SDMA, FDMA, TDMA, and CDMA.

CO2: To evaluate various wireless telecommunications systems.

CO3: To understand satellite communication systems, mobile IP challenges, and routing protocols in ad-hoc and sensor networks.

CO4: To investigate the impact of mobility on file systems, databases, web access, and application layers, including technologies like WAP.

References:

1. J. Schiller, Mobile Communications, 2nd edition, Addison Wesley.
2. William Stallings, Wireless Communications and Networks, 2nd edition, Prentice Hall
3. M. V. D. Heijden, M. Taylor, Understanding WAP, Artech House.
4. Charles Perkins, Mobile IP, Addison Wesley.
5. Charles Perkins, Ad hoc Networks, Addison Wesley.

Master of Technology

Wireless Mobile Networks Lab (MCSE-252)

Credits: 0+2

Tools: a. Wireshark b. Cisco Packet Tracer c. NS-2 / NS-3 d. OmNet++, OverSim

Lab Exercises:

1. The above mentioned tools can be used to experiment with Wireless Network.
2. Different protocols can be implemented and tested using the tools viz. AODV, DSR, DSDV etc.
3. Wireless topologies (WLAN) can be created for analysis and debugging purpose.
4. Test-beds can be developed for testing and performance analysis purposes.

Note: Implementation may be done in C, C++, Java and Python also. Experiments may be added or removed in justified manner.

Master of Technology

Blockchain and Its Applications (MCSE-203)

Credits: 3+0

Course Objectives: Student will be able to share information amongst all parties that access it via an application. Access to this ledger in terms of reading and writing may be unrestricted ('permissionless'), or restricted ('permissioned').

Syllabus

1. Introduction to Blockchain: The growth of blockchain technology - Progress toward maturity
Increasing interest, Distributed systems, the history of blockchain and Bitcoin - The events that led to blockchain, Electronic cash, Blockchain - Blockchain defined, Blockchain architecture, Generic elements of a blockchain, Benefits, features, and limitations of blockchain, Types of blockchain. Consensus: Consensus mechanism, Types of consensus mechanisms, Consensus in blockchain, CAP theorem and blockchain.
2. Decentralization: Decentralization using blockchain, Methods of decentralization - Disintermediation, Contest-driven decentralization, Routes to decentralization - How to decentralize, Decentralization framework example, Blockchain and full ecosystem decentralization - Storage, Communication, Computing power and decentralization, Pertinent terminology- Smart contracts, Autonomous agents, Decentralized organizations, Decentralized autonomous organizations, Decentralized autonomous corporations, Decentralized autonomous societies, Decentralized applications. Cryptographic constructs and blockchain technology: Homomorphic encryption, Signcryption, Secret sharing, Commitment schemes, Zero-knowledge proofs, Different types of digital signatures, Encoding schemes, Applications of cryptographic hash functions.
3. Introducing the consensus problem- The Byzantine generals problem, Fault tolerance, State machine replication, FLP impossibility, Lower bounds on the number of processors to solve consensus, Analysis and design - Model, Processes, Timing assumptions, Classification, Algorithms - CFT algorithms, BFT algorithms, Choosing an algorithm - Finality, Speed, performance, and scalability
4. Ethereum: An overview, The yellow paper, Ethereum - a user's perspective, The Ethereum network - The mainnet, Testnets, Private nets, Components of the Ethereum ecosystem - Keys and addresses, Accounts, Transactions and messages, Ether cryptocurrency/tokens (ETC and ETH), The Ethereum Virtual Machine (EVM) - Execution environment, The machine state, The iterator function, Smart contracts - Native contracts.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To comprehend the core concepts of blockchain technology and recognize its evolution and significance.

CO2: To identify the essential elements of a blockchain and differentiate between permissionless and permissioned consensus models.

CO3: To gain proficiency in creating smart contracts using Solidity, understand Decentralized Autonomous Organizations (DAOs), and manage smart contract interactions and inheritances.

CO4: To Investigate decentralized identity management, blockchain interoperability, and various real-world applications of blockchain technology.

References:

1. Mastering Bitcoin: Unlocking Digital Cryptocurrencies, by Andreas Antonopoulos
2. Melanie Swa ,Blockchain, O'Reilly
3. Hyperledger, Fabric - <https://www.hyperledger.org/projects/fabric>
4. Zero to Blockchain - An IBM Redbooks course, by Bob Dill, David Smits -
<https://www.redbooks.ibm.com/Redbooks.nsf/RedbookAbstracts/crse0401.html>

Master of Technology

Advanced Distributed Systems (MCSE-204)

Credit: 3+0

Course Objectives: Students will be able to achieving high scalability, fault tolerance, improved performance, efficient resource management across multiple nodes, handling complex distributed data consistency, implementing sophisticated communication protocols, designing self-healing mechanisms, and exploring advanced concepts like distributed consensus algorithms, large-scale data processing, and real-time distributed applications, all while maintaining user transparency and addressing potential challenges like network latency and security concerns.

Syllabus

1. Fundamentals of Distributed Systems: Introduction to Distributed Systems ,Definition, characteristics, and goals,Architectural models: client-server, peer-to-peer, micro services, Challenges: scalability, fault tolerance, security,Communication in Distributed Systems ,Remote Procedure Call (RPC),Remote Method Invocation (RMI),Message Passing and Publish-Subscribe Model,gRPC and REST APIs in distributed applications
2. Synchronization and Consistency:Time Synchronization ,Logical and physical clocks (Lamport & Vector Clocks),Berkeley and NTP synchronization,Consistency and Replication ,Data consistency models: strong, eventual, causal,Distributed shared memory,CAP theorem and its implications,Consensus Algorithms ,Paxos, Raft, Byzantine Fault Tolerance (BFT),Leader election in distributed systems
3. Fault Tolerance and Scalability:Failure Models & Fault Tolerance Techniques ,Crash faults, omission faults, Byzantine faults,Checkpointing and rollback recovery,Distributed Storage and Databases Distributed file systems (Google File System, HDFS),NoSQL databases (Cassandra, MongoDB, DynamoDB),Replication strategies and partitioning,Scalability in Distributed Systems ,Load balancing techniques
4. Security, Blockchain & Emerging Trends: Security in Distributed Systems ,Authentication, authorization, encryption,Secure multiparty computation,Distributed Denial of Service (DDoS) mitigation, Blockchain basics,Sharding and layer -2 solutions.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To learn epidemic-based algorithms, peer-to-peer networks, and distributed hash tables (DHTs).

CO2: To explore logical clocks, mutual exclusion, leader election, and distributed spanning tree algorithms.

CO3: To study consistency models, the CAP theorem, Paxos, Raft, and Byzantine fault tolerance.

CO4: To develop rule generation methods and hybridize fuzzy and rough set techniques for advanced problem-solving.

References:

1. G. F. Coulouris, J. Dollimore and T. Kindberg, Distributed Systems: Concepts and Design, 4th Ed, Addison-Wesley, 2005.
2. S. Mullender (Ed), Distributed Systems, 2nd Ed, Addison-Wesley, 1994.
3. M. Singhal and N. Shrivastava, Advanced Concepts in Operating Systems, McGraw Hill, 1994.

Master of Technology

Advanced Distributed Systems Lab(MCSE-254)

Credits: 0+2

1. Implement concurrent day-time client-server application.
2. Configure following options on server socket and tests them: SO_KEEPALIVE, SO_LINGER, SO_SNDBUF, SO_RCVBUF, TCP_NODELAY
3. Simulate the functioning of Lamport"s Logical Clock in C.
4. Simulate the Distributed Mutual Exclusion in C.
5. Implement a Distributed Chat Server using TCP Sockets in C.
6. Simulate Balanced Sliding Window Protocol in C.

Note: Implementation may be done in C, C++, Java and Python also. Experiments may be added or removed in justified manner.

Master of Technology

Big Data and Its Applications (MCSE-205)

Credits: 3+0

Course Objectives: Students will be able to improve decision-making, increase efficiency, and gain insights from large amounts of data.

Syllabus

1. Getting an overview of Big Data: Introduction to Big Data, Structuring Big Data, Types of Data, Elements of Big Data, Big Data Analytics, and Advantages of Big Data Analytics. Introducing Technologies for Handling Big Data: Distributed and Parallel Computing for Big Data, Cloud Computing and Big Data, Features of Cloud Computing, Cloud Deployment Models, Cloud Services for Big Data, Cloud Providers in Big Data Market.
2. Understanding Hadoop Ecosystem: Introducing Hadoop, HDFS and MapReduce, Hadoop functions, Hadoop Ecosystem. Hadoop Distributed File System- HDFS Architecture, Concept of Blocks in HDFS Architecture, Namenodes and Datanodes, Features of HDFS. MapReduce. Introducing HBase- HBase Architecture, Regions, Storing Big Data with HBase, Combining HBase and HDFS, Features of HBase, Hive, Pig and Pig Latin, Sqoop, ZooKeeper, Flume, Oozie.
3. Understanding MapReduce Fundamentals and HBase: The MapReduceFramework ,Exploring the features of MapReduce, Working of MapReduce, Techniques to optimize MapReduce Jobs, Hardware/Network Topology, Synchronization, File system, Uses of MapReduce, Role of HBase in Big Data Processing- Characteristics of HBase. Understanding Big Data Technology Foundations: Exploring the Big Data Stack, Data Sources Layer, Ingestion Layer, Storage Layer, Physical Infrastructure Layer, Platform Management Layer, Security Layer, Monitoring Layer, Visualization Layer.
4. Storing Data in Databases and Data Warehouses: RDBMS and Big Data, Issues with Relational Model, Non – Relational Database, Issues with Non-Relational Database, Polyglot Persistence, Integrating Big Data with Traditional Data Warehouse, Big Data Analysis and Data Warehouse.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To grasp the fundamentals of Big Data, Hadoop's architecture, and its core components.

CO2: To apply techniques for data ingestion and perform Extract, Transform, Load (ETL) operations using Hadoop tools.

CO3: To employ distributed processing paradigms like MapReduce and Pig for efficient data analysis.

CO4: To leverage functional programming in Scala and utilize Apache Spark for advanced data processing tasks.

References:

1. Abraham Silberschartz, Henry F. Korth and S. Surarshan ,Database system concepts :, McGraw Hill, 2011.
2. Quick-Start Guide: Learn the Essentials of Big Data Computing in the Apache Hadoop
3. Ecosystem, Douglas Eadline, Addison-Wesley, Pearson Education India; First edition 1 March 2016.
4. Nathan Marz, James Warren,Big Data: Principles and Best Practices of Scalable Real-Time Data Systems, Simon and Schuster, 2015

Master of Technology

Advanced Computer Networks (MCSE-206)

Credit: 3+0

Course Objectives: Students will be able to provide a deep understanding of complex and cutting-edge network technologies, including advanced routing algorithms, network performance optimization, security protocols, software-defined networking (SDN), cloud networking, and other emerging trends, allowing students to analyze, design, and implement highly efficient and secure network systems for modern computing environments.

Syllabus

1. **COMPUTER NETWORKS AND THE INTERNET** Computer Networks and the Internet: History of Computer Networking and the Internet, Networking Devices, The Network edge, The Network core, Access Networks and Physical media, ISPs and Internet Backbones. Networking Models: 5-layer TCP/IP Model, 7-Layer OSI Model, Internet Protocols and Addressing, Equal- Sized Packets Model: ATM.
2. **NETWORK ROUTING ROUTING AND ITS CONCEPTS** Network Routing Routing and its concepts: Structure of a Router, Basic Router Configuration, Building a Routing Table, Static Routing, Dynamic Routing Distance Vector Routing Protocol (RIPv1, RIPv2, EIGRP), Link State Routing Protocols (OSPF).
3. **LAN SWITCHING: SWITCHING AND ITS CONCEPTS** LAN Switching: Switching and its concepts: Structure of a Switch, Basic Switch Configuration, Virtual LANs (VLANs), VLAN Trunking Protocol (VTP), Spanning Tree Protocol (STP), Inter-VLAN Routing.
4. **WIDE AREA NETWORKS (WANS)** Wide Area Networks (WANs): Introduction to WANs, Point-to-Point Protocol (PPP) concepts, Frame Relay concepts, Dynamic Host Configuration Protocol (DHCP), Network Address Translation (NAT), IPv6. **NETWORK PROGRAMMING USING JAVA** M.Tech-CSE- Academic Regulations & Syllabus MLR18MLR Institute of Technology Page 42 Network Programming using Java: TCP sockets, UDP sockets (datagram sockets), Server programs that can handle one connection at a time and multiple connections (using multithreaded server), Remote Method Invocation (Java RMI) - Basic RMI Process, Implementation details Client Server Application.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To explain the history, architecture, and components of computer networks, including the OSI and TCP/IP models.

CO2: To demonstrate proficiency in configuring routers and switches, implementing VLANs, and managing network protocols.

CO3: To apply knowledge of routing algorithms and protocols such as RIP, OSPF, and EIGRP to design efficient network routing strategies.

CO4: To create network applications using Java, including TCP and UDP socket programming, and implement client-server models.

References:

1. Computer Networking: A Top-Down Approach Featuring the Internet, James F. Kurose, Keith W. Ross, Fifth Edition, Pearson Education, 2012.
2. Network Fundamentals, Mark Dye, Pearson Education.

3. A Systems approach, Larry L. Peterson & Bruce S. Davie, Fifth edition, Elsevier, 2012. 2. Computer Networks: A Top-Down Approach, Behrouz A. Forouzan, Firoz Mosharaf, Tata McGraw Hill, 2012.
4. Andrew S. Tanenbaum, Computer Networks, PHI

**Professional Elective Courses offered by
Department of Computer Science and Engineering**

Professional Elective Course List-1

Master of Technology

Sensor Network (MCSEE -101)

Credit: 3+0

Course Objectives: It focuses on designing efficient, scalable, and secure sensor networks for real-time monitoring in various domains such as healthcare, agriculture, and disaster management.

Syllabus

1. Basics of wireless sensor networks Some Foundational Information of WSN, Next-Generation Sensor Networked Tiny Devices Sensor Network Software, Unique Characteristics of Programming Environments for Sensor Network Future Demands on Sensor-Based Software Sensor Network Applications Characteristics of Sensor Networks, Nature of Data in Sensor Networks Overview of IEEE 802.15.4 and ZigBee.
2. Medium Access Control Protocols for Wireless Sensor Networks Introduction, Background Fundamentals of MAC Protocols MAC Protocols for WSNs: Schedule-Based Protocols and Random Access-Based Protocols Sensor-MAC: Protocol Overview, Periodic Listen and Sleep Operations, Schedule Selection and Coordination, Schedule Synchronization, Adaptive Listening, Access Control and Data Exchange (B-MAC,H-MAC, IMAC, O-MAC, S-MAC. Ri-MAC, T-MAC, Q-MAC (Query MAC), Q-MAC (QoS MAC), XMAC)
3. Routing Protocols for Wireless Sensor Networks Introduction, Data Dissemination and Gathering Routing Challenges and Design Issues in Wireless Sensor Networks, Network Scale and Time-Varying Characteristics, Resource Constraints Sensor Applications Data Models Routing Strategies in Wireless Sensor Networks: WSN Routing Techniques Flooding and Its Variants Sensor Protocols for Information via Negotiation Low-Energy Adaptive Clustering Hierarchy Power-Efficient Gathering in Sensor Information Systems Directed Diffusion, Geographical Routing
4. Transport Control Protocols and Operating Systems for WSN Traditional Transport Control Protocols: TCP (RFC 793), UDP (RFC 768), Mobile IP Feasibility of Using TCP or UDP for WSNs Introduction of Operating Systems Examples of Operating Systems: Tiny OS, Mate, Magnet OS

Course Outcomes (CO): At the end of this course students will be able:

CO1: To gain foundational knowledge of WSN architecture, characteristics, applications, and communication protocols like IEEE 802.15.4 and ZigBee.

CO2: To learn and analyze various MAC protocols (B-MAC, S-MAC, XMAC, etc.) and routing techniques (Flooding, LEACH, Directed Diffusion, Geographical Routing) used in WSNs.

CO3: To evaluate the feasibility of traditional transport protocols (TCP, UDP) for WSNs and study specialized operating systems like TinyOS, Mate, and MagnetOS.

CO4: To develop the ability to design and optimize WSN-based applications considering energy efficiency, scalability, and real-time data processes requirements.

References:

1. Algorithms for Sensor and Ad Hoc Networks, Advanced Lectures, Lecture Notes in Computer Science 4621, Editors Dorothea Wagner and Roger Wattenhofer, 2007.

2. Fundamentals of Wireless Sensor Networks: Theory and Practice Waltenegus Dargie, Christian Poellabauer John Wiley & Sons, 2010
3. Ad Hoc and Sensor Networks: Theory and Applications Carlos De Moraes Cordeiro, Dharma Prakash Agrawal World Scientific, 2011.

Master of Technology

Software Testing and Auditing (MCSEE -102)

Credit: 3+0

Course Objectives: The Software Testing and Auditing aims to equip students with knowledge and skills in software quality assurance, testing methodologies, and software auditing practices.

Syllabus

1. **TESTING BASICS** Testing as an engineering activity – Role of process in software quality – Testing as a process – Basic definitions – Software testing principles – The tester's role in a software development organization – Origins of defects – Defect classes – The defect repository and test design – Defect examples – Developer / Tester support for developing a defect repository.
2. **TEST CASE DESIGN** Introduction to testing design strategies – The smarter tester – Test case design strategies – Using black box approach to test case design – Random testing – Equivalence class partitioning – Boundary value analysis – Other black box test design approaches – Black box testing and COTS – Using white box approach to test design – Test adequacy criteria – Coverage and control flow graphs – Covering code logic – Paths – Their role in white box based test design – Additional white box test design approaches – Evaluating test adequacy criteria.
3. **LEVELS OF TESTING** The need for levels of testing – Unit test – Unit test planning – Designing the unit tests – The class as a testable unit – The test harness – Running the unit tests and recording results – Integration tests – Designing integration tests – Integration test planning – System test – The different types – Regression testing – Alpha, beta and acceptance tests.
4. **TEST MANAGEMENT -CONTROLLING AND MONITORING** Basic concepts – Testing, debugging goals, policies – Test planning – Test plan components – Test plan attachments – Locating test items – Reporting test results – The role of three groups in test planning and policy development – Process and the engineering disciplines – Introducing the test specialist – Skills needed by a test specialist – Building a testing group. Defining terms – Measurements and milestones for controlling and monitoring – Status meetings – Reports and control issues – Criteria for test completion – SCM – Types of reviews – Developing a review program – Components of review plans – Reporting review results.

Program Outcomes (PO): At the end of this course students will be able

C01: To gain knowledge of software testing as an engineering activity.

C02: To learn and apply black-box and white-box testing approaches, test adequacy criteria, and effective test case design strategies to enhance software reliability..

C03: To develop skills in unit, integration, system, regression, and acceptance testing, along with planning, execution, and result analysis for different testing phases.

C04: To acquire knowledge in test planning, debugging, reporting, review processes, and test team management to ensure systematic and efficient software testing.

References:

1. SrinivasanDesikan, Gopaldaswamy Ramesh, "Software Testing: Principles and Practices", Pearson 2012
2. Aditya P. Mathur, "Foundations of Software Testing", Pearson, 2008
3. Paul Ammann, Jeff Offutt, "Introduction to Software Testing", Cambridge University Press, 2008
4. Paul C. Jorgensen, "Software Testing: A Craftsman's Approach", Auerbach Publications, 2008

Master of Technology

Real Time Systems (MCSEE -103)

Credit: 3+0

Course Objectives: The Real-Time Systems course aims to provide students with a comprehensive understanding of time-constrained computing environments, their design principles, and implementation techniques.

Syllabus

1. Introduction: Definition, Typical Real Time Applications: Digital Control, High Level Controls, Signal Processing etc., Release Times, Deadlines, and Timing Constraints, Hard Real Time Systems and Soft Real Time Systems, Reference Models for Real Time Systems: Processors and Resources, Temporal Parameters of Real Time Workload, Periodic Task Model, Precedence Constraints and Data Dependency.

2. Real Time Scheduling Common Approaches to Real Time Scheduling: Clock Driven Approach, Weighted Round Robin Approach, Priority Driven Approach, Dynamic Versus Static Systems, Optimality of Effective-Deadline-First (EDF) and Least-Slack-Time-First (LST) Algorithms, Rate Monotonic Algorithm, Offline Versus Online Scheduling, Scheduling Aperiodic and Sporadic jobs in Priority Driven and Clock Driven Systems.

3. Resources Sharing: Effect of Resource Contention and Resource Access Control (RAC), Nonpreemptive Critical Sections, Basic Priority-Inheritance and Priority-Ceiling Protocols, Stack Based Priority- Ceiling Protocol, Use of Priority-Ceiling Protocol in Dynamic Priority Systems, Preemption Ceiling Protocol, Access Control in Multiple-Unit Resources, Controlling Concurrent Accesses to Data Objects.

4. Real Time Communication Basic Concepts in Real time Communication, Soft and Hard RT Communication systems, Model of Real Time Communication, Priority-Based Service and Weighted Round-Robin Service Disciplines for Switched Networks, Medium Access Control Protocols for Broadcast Networks, Internet and Resource Reservation Protocols.

Course Outcomes (CO): At the end of this course students will be able:

CO1: To gain knowledge of software testing as an engineering activity, defect identification, defect management, and the role of testing in ensuring software quality

CO2: To learn and apply black-box and white-box testing approaches, test adequacy criteria, and effective test case design strategies to enhance software reliability..

CO3: To understand resource contention issues, priority-inheritance and priority-ceiling protocols, and techniques for controlling concurrent access to shared resources.

CO4: To gain knowledge of real-time communication models, service disciplines, MAC protocols, and resource reservation techniques for ensuring reliable communication in real-time environments.

References:

- 1.** Real Time Systems by Jane W. S. Liu, Pearson Education Publication.
- 2.** Mall Rajib, "Real Time Systems", Pearson Education

3. Albert M. K. Cheng , “Real-Time Systems: Scheduling, Analysis, and Verification”, Wiley.

Master of Technology

Genetic Algorithms (MCSEE -104)

Credit: 3+0

Course Objectives: The Genetic Algorithms (GA) course aims to provide students with a deep understanding of evolutionary computation techniques used for solving optimization and search problems.

Syllabus

1. Introduction: Robustness of Traditional Optimization and Search Methods, The goals of Optimization, How are Genetic Algorithms Different from Traditional Methods?, A simple genetic algorithm, Genetic algorithms at work – a simulation by hand, Grist for the Search Mill – important similarities, Similarity templates (Schemata), Learning the Lingo.
2. Genetic Algorithms Revisited: Mathematical Foundations – Who shall live and who shall die? The fundamental theorem, Schema Processing at work: An example by hand revisited, the twoarmed and K-armed bandit problem, How many schemata are processed usefully?, The building block hypothesis, another perspective: the minimal Deceptive problem, Schemata revisit: similarity templates as hyper planes.
3. Computer Implementation Of A Genetic Algorithm – Data structures, reproduction, crossover, and mutation, A time to reproduce, a time to cross, get with the main program, How well does it work? Mapping objective functions to fitness form, fitness scaling, codings, a multiparameter, mapped, fixed-point coding, discretization, Constraint Handling.
4. Techniques In Genetic Search – Dominance, diploidy and abeyance, inversion and other reordering operators, other micro operators, niche and speciation, multiobjective optimization - Knowledge-based techniques, genetic algorithms and parallel processors.

Course Outcomes (CO): At the end of this course students will be able

CO1: To learn the principles of genetic algorithms, their differences from traditional optimization methods, and their role in solving complex search problems..

CO2: To gain insights into mathematical foundations, schema processing, the building block hypothesis, and the impact of selection, crossover, and mutation.

CO3: To develop skills in designing and coding genetic algorithms, handling constraints, fitness scaling, and mapping objective functions for optimization tasks.

CO4: To understand multi-objective optimization, dominance, niche formation, and parallel processing techniques to enhance genetic search efficiency.

References:

1. David E.Goldberg, "Genetic Algorithms" – 1/e, Pearson Education.
2. Genetic algorithms in search, optimization and Mechine learning, By David E. Gold Berg Pearson Edition
3. An Introduction to Genetic Algorithm by Melanie Mitchell
4. The Simple Genetic Algorithm Foundation & Theores by Michael P. Vosk

Master of Technology

Neural Networks (MCSEE-105)

Credit: 3+0

Course Objectives: The Neural Networks course aims to provide students with a strong foundation in artificial neural networks (ANNs), their architecture, learning techniques, and real-world applications.

Syllabus

1. Basics of ANN: Models to Neuron; Basic learning laws. Activation and Synaptic Dynamics: Activation dynamics models; Synaptic dynamics models; Stability and Convergence.
2. Analysis of Feed forward Neural Networks: Linear associative networks for pattern association; Single layer and Multilayer Perception network for pattern classification; Multilayer feed forward neural networks for pattern mapping.
3. Analysis of Feedback Neural Networks: Linear auto associative networks; Hopfield model for pattern storage; stochastic networks; Boltzmann machine for pattern environment storage.
4. Competitive Learning Neural Networks: Basic competitive learning laws; Analysis of pattern clustering networks; Analysis of self-organizing feature mapping networks. Applications of ANN: Pattern classification problems; Optimization; Control.

Course Outcomes (CO): At the end of this course students will be able

CO1: To learn the fundamental neuron models, learning laws, activation, and synaptic dynamics, along with stability and convergence concepts.

CO2: To gain expertise in linear associative networks, perceptron models, and multilayer feedforward networks for pattern classification and mapping.

CO3: To understand auto-associative networks, Hopfield models, and Boltzmann machines for pattern storage and stochastic learning.

CO4: To learn competitive learning laws, self-organizing feature maps, and real-world applications in pattern classification, optimization, and control.

References:

1. J.A. Anderson, An Introduction to Neural Networks, MIT
2. Hagen Demuth Beale, Neural Network Design, Cengage Learning
3. Laurene V. Fausett, "Fundamentals of Neural Networks : Architectures, Algorithms and Applications", Pearson India
4. Kosko, Neural Network and Fuzzy Sets, PHI

Master of Technology

Data Warehousing & Data Mining (MCSEE-106)

Credit: 3+0

Course Objectives: The Data Warehousing & Data Mining course aims to provide students with a strong foundation in data storage, management, and analysis techniques to support decision-making.

Syllabus

1. Data Warehousing and Business Analysis: - Data warehousing Components, Building a Data warehouse, Mapping the Data Warehouse to a Multiprocessor Architecture, DBMS Schemas for Decision Support, Data Extraction, Cleanup, and Transformation Tools, Metadata reporting, Query tools and Applications, Online Analytical Processing (OLAP) – OLAP and Multidimensional Data Analysis.
2. Data Mining: - Data Mining Functionalities – Data Preprocessing, Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization and Concept Hierarchy Generation. Association Rule Mining: - Efficient and Scalable Frequent Item set Mining Methods, Mining Various Kinds of Association Rules, Association Mining to Correlation Analysis, ConstraintBased Association Mining.
3. Classification and Prediction: - Issues Regarding Classification and Prediction, Classification by Decision Tree Introduction, Bayesian Classification, Rule Based Classification, Classification by Back propagation, Support Vector Machines, Associative Classification, Lazy Learners, Other Classification Methods, Prediction Accuracy and Error Measures, Evaluating the Accuracy of a Classifier or Predictor , Ensemble Methods, Model Section.
4. Cluster Analysis: - Types of Data in Cluster Analysis, A Categorization of Major Clustering Methods, Partitioning Methods, Hierarchical methods, Density-Based Methods. Grid-Based Methods, Model-Based Clustering Methods, Clustering High-Dimensional Data, ConstraintBased Cluster Analysis, Outlier Analysis.

Course Outcomes (CO): At the end of this course students will be able

CO1: To gain knowledge of data warehousing components.

CO2: To learn data preprocessing techniques, association rule mining, and scalable algorithms for discovering patterns and correlations in large datasets.

CO3: To develop expertise in classification techniques such as decision trees, Bayesian classification, SVMs, and ensemble methods for predictive analytics

CO4: To understand various clustering techniques, including partitioning, hierarchical, density-based, and grid-based methods, for effective pattern recognition and anomaly detection.

References:

1. Jiawei Han and Micheline Kamber “Data Mining Concepts and Techniques” Second Edition,
2. Alex Berson and Stephen J. Smith “Data Warehousing, Data Mining & OLAP”, Tata McGraw – Hill Edition, Tenth Reprint 2007.

3. G. K. Gupta "Introduction to Data Mining with Case Studies", Easter Economy Edition, Prentice Hall of India, 2006.
4. Pang-Ning Tan, Michael Steinbach and Vipin Kumar "Introduction to Data Mining", Pearson Education, 2007.
5. Soman K.P., Shyam Diwakar and V. Ajay, "Insight into Data mining Theory and Practice", Easter Economy Edition, Prentice Hall of India, 2006.
6. Daniel T.Larose, "Data Mining Methods and Models", Wile-Interscience, 2006. GENETIC ALGORITHM

Professional Elective Course List-2

Master of Technology

Machine Learning (MCSEE-201)

Credit: 3+0

Course Objectives: The Machine Learning (ML) course aims to provide students with a comprehensive understanding of ML concepts, algorithms, and applications for data-driven decision-making.

Syllabus

1. INTRODUCTION – Well defined learning problems, Designing a Learning System, Issues in Machine Learning; THE CONCEPT LEARNING TASK - General-to-specific ordering of hypotheses, Find-S, List then eliminate algorithm, Candidate elimination algorithm, Inductive bias
2. DECISION TREE LEARNING - Decision tree learning algorithm-Inductive bias- Issues in Decision tree learning; ARTIFICIAL NEURAL NETWORKS – Perceptrons, Gradient descent and the Delta rule, Adaline, Multilayer networks, Derivation of backpropagation rule Backpropagation Algorithm Convergence, Generalization.
3. EVALUATING HYPOTHESES – Estimating Hypotheses Accuracy, Basics of sampling Theory, Comparing Learning Algorithms; BAYESIAN LEARNING – Bayes theorem, Concept learning, Bayes Optimal Classifier, Naïve Bayes classifier, Bayesian belief networks, EM algorithm.
4. COMPUTATIONAL LEARNING THEORY – Sample Complexity for Finite Hypothesis spaces, Sample Complexity for Infinite Hypothesis spaces, The Mistake Bound Model of Learning; INSTANCE-BASED LEARNING – k-Nearest Neighbor Learning, Locally Weighted Regression, Radial basis function networks, Case-based learning GENETIC ALGORITHMS – an illustrative example, Hypothesis space search, Genetic Programming, Models of Evolution and Learning; Learning first order rules-sequential covering algorithms-General to specific beam search-FOIL; REINFORCEMENT LEARNING - The Learning Task, Q Learning.

Course Outcomes (CO): At the end of this course students will be able

CO1: To gain knowledge of well-defined learning problems, hypothesis space, concept learning algorithms, and inductive bias.

CO2: To develop expertise in decision tree learning, artificial neural networks (ANN), backpropagation, Bayesian learning, and hypothesis evaluation techniques.

CO3: To understand computational learning theory, instance-based learning (k-NN, RBF networks), genetic algorithms, and evolutionary learning models.

CO4: To learn reinforcement learning techniques like Q-learning and rule-learning approaches such as sequential covering and beam search.

References:

1. Tom.M.Mitchell, Machine Learning, McGraw Hill International Edition
2. Ethern Alpaydin, Introduction to Machine Learning. Eastern Economy Edition, Prentice Hall of India, 2005.
3. Bishop, C., Pattern Recognition and Machine Learning. Berlin: Springer-Verlag.

Master of Technology

High Performance Networking (MCSEE-202)

Credit: 3+0

Course Objectives: The High Performance Networking (HPN) course aims to provide students with a deep understanding of advanced networking architectures, protocols, and optimization techniques to support high-speed, low-latency communication systems.

Syllabus

1. Switching Networks: Switching – Packet switching - Ethernet, Token Ring, FDDI, DQDB, Frame Relay, SMDS, Circuit Switched – SONET, DWDM, DSL, Intelligent Networks – CATV, ATM – Features, Addressing, Signaling & Routing, Header Structure, ATM Adaptation layer, Management control, BISDN, Internetworking with ATM.
2. Multimedia Networking Applications : Streaming stored Audio and Video, Best effort service, protocols for real time interactive applications, Beyond best effort, scheduling and policing mechanism, integrated services, RSVP- differentiated services.
3. Advanced Networks Concepts: VPN-Remote-Access VPN, site-to-site VPN, Tunneling to PPP, Security in VPN. MPLS operation, Routing, Tunneling and use of FEC, Traffic Engineering, and MPLS based VPN, overlay networks P2P connections. -IPv4 vs V6.
4. Packet Queues And Delay Analysis: Little's theorem, Birth and Death process, queueing discipline- Control & stability -, Markovian FIFO queueing system, Non-markovian - Pollaczek-Khinchin formula and M/G/1, M/D/1, self-similar models and Batch-arrival model, Networks of Queues – Burke's theorem and Jackson Theorem. Network Security And Management: Principles of cryptography – Elliptic-AES Authentication – integrity – key distribution and certification– Access control and: fire walls – DoS-attacks and counter measures – security in many layers. Infrastructure for network management – The internet standard management framework – SMI, MIB, SNMP, Security and administration – ASN.1.

Course Outcomes (CO): At the end of this course students will be able

CO1: To gain knowledge of packet and circuit-switched networks, including Ethernet, ATM, SONET, DSL, and their addressing, signaling, and adaptation layers.

CO2: To learn about real-time streaming, best-effort services, scheduling mechanisms, and protocols like RSVP and differentiated services for multimedia applications.

CO3: To understand VPNs, MPLS, tunneling, traffic engineering, P2P connections, and IPv4 vs. IPv6 for efficient and secure network communication.

CO4: To study queueing models, cryptographic principles, authentication, firewalls, DoS countermeasures, and network management frameworks like SNMP and ASN.1.

References:

1. Aunurag Kumar, D. Manjunath, Joy Kuri, "Communication Networking", Morgan Kaufmann Publishers, 2011.
2. J.F. Kurose & K.W. Ross, "Computer Networking- A Top Down Approach Featuring the Internet", Pearson, 2nd Edition, 2003.
3. Nader F. Mir, "Computer and Communication Networks", Pearson Education, 2009.
4. Walrand J. Varatya, "High Performance Communication Network", Morgan Kaufmann – Harcourt Asia Pvt. Ltd., 2nd Edition, 2000.

5. Hersent Gurle & petit, "IP Telephony, Packet Pored Multimedia Communication Systems", Pearson Education 2003.
6. Fred Halsall and Lingana Gouda Kulkarni, "Computer Networking and the Internet", Fifth Edition, Pearson Education, 2012.
7. Larry L.Peterson & Bruce S.David, "Computer Networks: A System Approach"- Morgan Kaufmann Publisher, 1996.

Master of Technology

Big Data Analytics (MCSEE-203)

Credit: 3+0

Course Objectives: The Big Data Analytics course aims to provide students with a comprehensive understanding of big data technologies, tools, and methodologies for analyzing large-scale datasets to extract meaningful insights

Syllabus

1. **INTRODUCTION TO BIG DATA:** Big Data and its Importance, Four V's of Big Data, Drivers for Big Data, Introduction to Big Data Analytics, Big Data Analytics applications. **BIG DATA TECHNOLOGIES:** Hadoop's Parallel World, Data discovery, Open source technology for Big Data Analytics, cloud and Big Data, Predictive Analytics, Mobile Business Intelligence and Big Data, Crowd Sourcing Analytics, Inter- and Trans-Firewall Analytics, Information Management.
2. **PROCESSING BIG DATA:** Integrating disparate data stores, Mapping data to the programming framework, Connecting and extracting data from storage, Transforming data for processing, subdividing data in preparation for Hadoop Map Reduce.
3. **HADOOP MAPREDUCE:** Employing Hadoop Map Reduce, Creating the components of Hadoop Map Reduce jobs, Distributing data processing across server farms, Executing Hadoop Map Reduce jobs, monitoring the progress of job flows, The Building Blocks of Hadoop Map Reduce Distinguishing Hadoop daemons, Investigating the Hadoop Distributed File System Selecting appropriate execution modes: local, pseudo-distributed, fully distributed.
4. **BIG DATA TOOLS AND TECHNIQUES:** Installing and Running Pig, Comparison with Databases, Pig Latin, User- Define Functions, Data Processing Operators, Installing and Running Hive, Hive QL, Querying Data, User-Defined Functions, Oracle Big Data.

Course Outcomes (CO): At the end of this course students will be able

CO1: To gain knowledge of Big Data fundamentals, the Four V's (Volume, Velocity, Variety, and Veracity)

CO2: To learn techniques for integrating disparate data sources, mapping data to programming frameworks

CO3: To develop skills in creating, executing, and monitoring Hadoop MapReduce jobs, understanding the Hadoop Distributed File System (HDFS)

CO4: To gain practical experience in using tools like Pig and Hive for querying and processing Big Data, writing Pig Latin scripts, and working with HiveQL to analyze large datasets efficiently.

References:

1. Michael Minelli, Michehe Chambers, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Business", 1st Edition, Ambiga Dhiraj, Wiely CIO Series, 2013.
2. Arvind Sathi, "Big Data Analytics: Disruptive Technologies for Changing the Game", 1st Edition, IBM Corporation, 2012.1.
3. Rajaraman, A., Ullman, J. D., Mining of Massive Datasets, Cambridge University Press, United Kingdom, 2012

4. Berman, J.J., Principles of Big Data: Preparing, Sharing and Analyzing Complex Information, Morgan Kaufmann, 2014
5. Barlow, M., Real-Time Big Data Analytics: Emerging Architecture, O Reilly, 2013
6. Schonberger, V.M. , Kenneth Cukier, K., Big Data, John Murray Publishers, 2013
7. Bill Franks, "Taming the Big Data Tidal Wave: Finding Opportunities in Huge Data Streams with Advanced Analytics", 1st Edition, Wiley and SAS Business Series, 2012.

Master of Technology

Software Metrics and Quality Assurance (MCSEE-204)

Credit: 3+0

Course Objectives: The Software Metrics and Quality Assurance course aims to provide students with a deep understanding of software measurement techniques, quality assurance processes, and best practices to ensure high-quality software development.

Syllabus

1. What Is Software Quality: Quality: Popular Views, Quality Professional Views, Software Quality, Total Quality Management and Summary. Fundamentals Of Measurement Theory: Definition, Operational Definition, And Measurement, Level Of Measurement, Some Basic Measures, Reliability And Validity, Measurement Errors, Be Careful With Correlation, Criteria For Causality, Summary. Software Quality Metrics Overview: Product Quality Metrics, In Process Quality Metrics, Metrics for Software Maintenance, Examples For Metrics Programs, Collecting Software Engineering Data.
2. Applying The Seven Basic Quality Tools In Software Development: Ishikawa's Seven Basic Tools, Checklist, Pareto Diagram, Histogram, Run Charts, Scatter Diagram, Control Chart, Cause And Effect Diagram. The Rayleigh Model: Reliability Models, The Rayleigh Model Basic Assumptions, Implementation, Reliability And Predictive Validity.
3. Complexity Metrics And Models: Lines Of Code, Halstead's Software Science , Cyclomatic Complexity Syntactic Metrics, An Example Of Module Design Metrics In Practice .Metric And Lessons Learned For Object Oriented Projects: Object Oriented Concepts And Constructs, Design And Complexity Metrics, Productivity Metrics, Quality And Quality Management Metrics, Lessons Learned For object oriented Projects. Availability Metrics: Definition And Measurement Of System Availability, Reliability Availability And Defect Rate, Collecting Customer Outage Data For Quality Improvement, In Process Metrics For Outage And Availability .Conducting Software Project Assessment :Audit Ad Assessment , Software Process Maturity Assessment And Software Project Assessment , Software Process Assessment A Proposed Software Project Assessment Method.
4. Dos And Don'ts Of Software Process Improvement :Measuring Process Maturity, Measuring Process Capability, Staged Versus Continuous Debating Religion, Measuring Levels Is Not Enough, Establishing The Alignment Principle , Take Time Getting Faster, Keep it Simple Or Face Decomplexification, Measuring The Value Of Process Improvement , Measuring Process Compliance , Celebrate The Journey Not Just The Destination. Using Function Point Metrics to Measure Software Process Improvement: Software Process Improvement Sequences, Process Improvement Economies, Measuring Process Improvement at Activity Levels.

Course Outcomes (CO): At the end of this course students will be able

CO1: To gain knowledge of software quality principles, total quality management (TQM), and fundamentals of measurement theory, including reliability, validity, and causality in software engineering.

CO2: To learn to use Ishikawa's seven basic quality tools, Pareto diagrams, control charts, and the Rayleigh reliability model for software development and quality assessment.

C03: To understand complexity metrics like Lines of Code (LOC), Halstead's metrics, Cyclomatic Complexity, and object-oriented design metrics for productivity and quality management.

C04: To develop skills in process maturity assessment, software project evaluation, function point metrics, and best practices for continuous software process improvement.

References:

1. Stephen H Khan: Metrics and Models in Software Quality Engineering, Pearson 2nd edition 2013.
2. Norman E-Fentor and Share Lawrence Pflieger." Software Metrics". International Thomson Computer Press, 1997.
3. S.A.Kelkar,"Software quality and Testing, PHI Learning, Pvt, Ltd., New Delhi 2012.
4. Watts S Humphrey, "Managing the Software Process", Pearson Education Inc, 2008.
5. Mary Beth Chrissis, Mike Konrad and Sandy Shrum, "CMMI", Pearson Education(Singapore) Pte Ltd, 2003'
6. Philip B Crosby, " Quality is Free: The Art of Making Quality Certain ", Mass Market, 1992.